

DSA5203: Project 3 Technical Report

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1. Introduction

This project focuses on scene classification using convolutional neural networks (CNNs). The proposed system integrates several advanced components to enhance performance, including the Convolutional Block Attention Module (CBAM) for attention-based feature refinement, Spatial Pyramid Pooling (SPP) for multi-scale feature extraction, residual connections to improve optimization, and partial replacement of fully connected layers with Global Average Pooling (GAP) for improved efficiency. Additionally, an ensemble approach is employed to further boost classification accuracy. The overall architecture draws inspiration from recent works by Yee et al. (2022), Woo et al. (2018) and He et al. (2015).

2. Preprocessing and data augmentation

To enhance model accuracy and reduce the risk of overfitting, we implemented a comprehensive data preprocessing and augmentation pipeline. This project adopts multi-size training using two different scales, i.e., 180 x 180 and 224 x 224 (Yee et al., 2022). During training, each image is randomly resized to one of these two scales. Subsequently, a series of data augmentation techniques is applied, including random affine transformations, horizontal flipping, color jittering, and random erasing. These augmentations significantly improve the model's generalization capability. All images are also normalized using the standard ImageNet mean and standard deviation values.

For validation and testing, images are resized to a fixed scale of 224 x 224, and the same normalization parameters are applied to ensure consistency with the training pipeline. The training dataset is split into 80% training and 20% validation subsets using stratified sampling to preserve class distribution. To accommodate the multi-size input strategy and convert image class labels into the required numerical format, custom collate functions are used during data batching for both training and validation phases.

3. Construction of CNN model

Our CNN model consists of five convolutional blocks, each augmented with CBAM and residual connections, followed by SPP layer and fully connected (FC) classification layer with partial GAP.

3.1. Convolutional block

Each convolutional block contains a convolutional layer with a kernel size of 3×3, padding of 1, and stride of 1, capturing local spatial features while preserving spatial dimensions. This is followed by batch normalization to stabilize training, ReLU activation for non-linearity, and 2×2 max pooling to progressively reduce the spatial resolution and increase the network's receptive field.

3.2. Convolutional Block Attention Module (CBAM)

Each block is enhanced with a CBAM module (Woo et al., 2018), which sequentially applies channel and spatial attention to adaptively refine feature representations. Channel attention emphasizes informative channels by using global average pooling to compress spatial dimensions and passing the result to a two-layer FC network with ReLU and sigmoid activations to generate channel-wise attention weights. Spatial attention captures important spatial regions within feature maps by concatenating average-pooled and max-pooled features across the channel axis and applies a convolution and sigmoid activation to generate a spatial attention map. The input feature map is multiplied elementwise by both attention maps, allowing the network to focus on informative features in both channel and spatial domains.

3.3. Residual connections

Residual connections (He et al., 2015) are included to improve gradient flow and mitigate the vanishing gradient problem. As the number of feature channels increases between blocks, a 1×1 convolutional projection is used to align dimensions before residual addition. These skip connections enable the model to learn residual mappings, improving optimization and allowing deeper networks.

3.4. Spatial Pyramid Pooling (SPP)

An SPP layer (Yee et al., 2022) is applied after the final convolution block. SPP aggregates features at multiple scales using adaptive max pooling at each level, then flattened and concatenated the features across all levels. A 3-level pyramid with adaptive max pooling with 1×1 , 2×2 , and 4×4 grid sizes is used. This allows the network to encode spatial hierarchies and capture both local and global contextual information without requiring fixed input size.

3.5. Fully Connected (FC) Classification layer with Partial Global Average Pooling (GAP)

Following SPP, we apply a partial GAP operation (Yee et al., 2022) to compress the multi-scale features across the spatial dimensions, reducing the feature map to a compact vector representation. This is followed by a FC classification layer, consisting of a dropout layer with a rate of 0.5 for regularization and a linear layer for final class prediction. This design leverages the efficiency of GAP for spatial summarization while preserving the learning capacity of an FC layer.

3.6. Training configurations

Dynamic zero-padding is implemented to handle multi-sized inputs during training while maintaining batch processing efficiency. All images in a batch are padded to match the maximum height and width present in that batch, avoiding the information loss associated with resizing while preserving compatibility with convolutional and pooling operations.

The model is optimized using the AdamW optimizer, which decouples weight decay from the gradient updates to provide more effective regularization. A weight decay coefficient of 0.0001 is applied to improve generalization and reduce overfitting. The cross-entropy loss function with a label smoothing factor of 0.1 is used to prevent overconfidence in predictions and improves generalizability. An adaptive learning rate scheduler (ReduceLROnPlateau) is employed to adjust the learning rate based on validation performance. The learning rate is initialized at 0.0001 and is halved when validation accuracy does not improve for 5 consecutive epochs. This encourages better convergence and avoids premature stagnation in performance. Training is conducted over 100 epochs, and the model achieving the highest validation accuracy is saved as the final model.

4. Further improvement on model performance

The custom CNN model constructed and trained as described in Section 3 achieved a training accuracy of 96.2% and a validation accuracy of 85.0%. To further enhance classification performance and address generalization limitations, we integrated two fine-tuned pre-trained CNN architectures, MobileNetV2 and VGG16, into our framework. These models were combined using a soft-voting ensemble approach. Ensemble methods have been shown to boost performance by leveraging the complementary strengths of diverse model architectures (Yee et al., 2022).

4.1. Fine-tuning pre-trained MobileNetV2 and VGG16

We preserved the convolutional backbones of MobileNetV2 and VGG16 to leverage the representational power of their pre-trained weights, replacing the original classification layer with our

custom architecture. This includes SPP layer and classification layer with partial GAP, consistent with the design outlined in Section 3. The models are initialized with pre-trained weights and fine-tuned during training to adapt effectively to the scene classification task. This strategy retains the general feature extraction capabilities of the pre-trained CNNs while enabling domain-specific adaptation.

The training configurations are similar to Section 3.6 with minor adjustments to the parameters. This includes limiting the training to only 50 epochs and uses initial learning rate of 0.001 for MobileNetV2 and 0.0005 for VGG16. As before, the best-performing model on the validation set is selected as the final model.

4.2. Ensemble via Soft Voting

The three models, i.e., custom CNN, MobileNetV2 and VGG16, independently predicts class probabilities using softmax activation. The predicted probability vectors are averaged across the models, and the class with the highest averaged probability is selected as the final prediction. This soft-voting ensemble aggregates predictions from models with diverse architectural biases, improving overall generalization. Moreover, the generalization gap decreased from 11.2% to 7.0%, indicating better robustness and reduced overfitting.

5. Conclusion

This project presents a robust scene classification model by developing a custom CNN architecture that strategically integrates several advanced components such as CBAM, SPP, residual connections, and partial GAP. These components collectively strengthen spatial feature learning, improve attention to salient regions, and enhance the model's ability to capture both local textures and global scene context. A comprehensive preprocessing pipeline and regularization techniques also ensure effective generalization of the model. Further improvement in performance is achieved through an ensemble with fine-tuned MobileNetV2 and VGG16, demonstrating the value of architectural diversity and transfer learning.

6. References

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